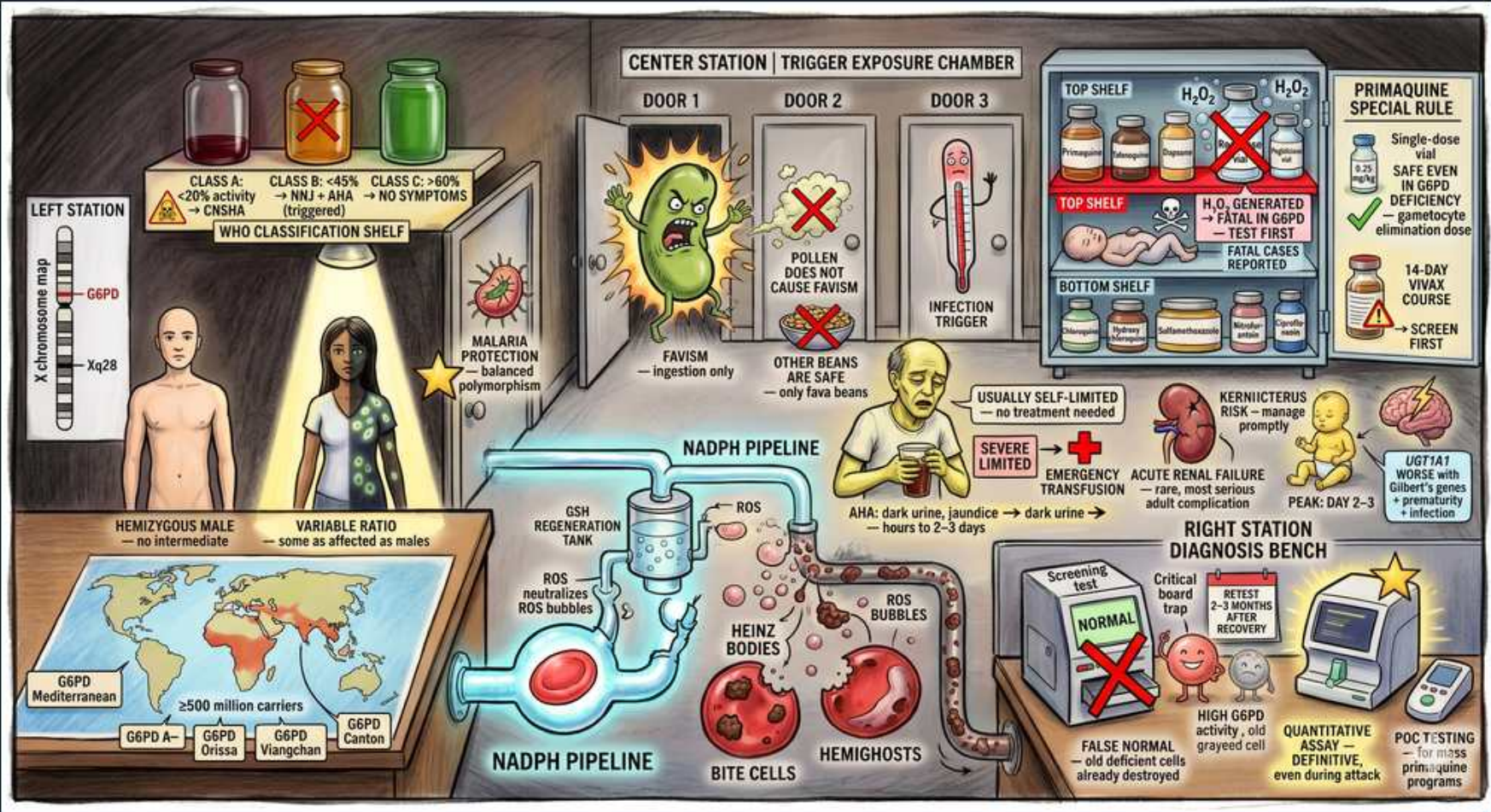


G6PD Deficiency — Classes, Triggers & Diagnosis



1. WHO classes by activity

WHAT YOU SEE

WHO classification shelf: Class A <20%, B 60%

WHAT IT ENCODES

G6PD severity is graded by enzyme activity: severe (chronic non-spherocytic hemolytic anemia, <10–20%), moderate Mediterranean-type with episodic hemolysis, and mild African A- variant. Activity level predicts whether hemolysis is chronic or only triggered.

BOARD TRAP

Class is by residual enzyme activity, not by ethnicity alone; the African A- variant is mild/episodic while Mediterranean is moderate-severe.

2. X-linked recessive (Xp28)

WHAT YOU SEE

Hemizygous male figure, Xp28 locus

WHAT IT ENCODES

G6PD is X-linked recessive (Xp28). Hemizygous males are fully affected; the trait is the most common human enzyme defect (~500 million carriers worldwide). Confers malaria protection — a balanced polymorphism.

BOARD TRAP

Don't call it autosomal — X-linkage explains male predominance and the variable female phenotype.

3. Variable female ratio (lyonization)

WHAT YOU SEE

Two-tone female figure, 'variable ratio'

WHAT IT ENCODES

Heterozygous females show variable severity due to random X-inactivation (lyonization), creating a mosaic of normal and deficient red cells. Some heterozygotes are as affected as males.

BOARD TRAP

A heterozygous female is NOT automatically protected — skewed lyonization can produce clinically significant hemolysis.

4. Geographic variants

WHAT YOU SEE

World map: G6PD A-, Mediterranean, Canton, Orissa

WHAT IT ENCODES

Common variants map to malaria-endemic regions: G6PD A- (African, mild), Mediterranean (severe favism), and Asian variants (Canton, Viangchan). Geography predicts severity and trigger sensitivity.

BOARD TRAP

Mediterranean variant patients get severe favism; African A- patients usually do not — variant identity changes management.

5. Oxidative drug triggers

WHAT YOU SEE

Top-shelf high-risk drug jars, H₂O₂ generated

WHAT IT ENCODES

Oxidant drugs deplete NADPH and overwhelm glutathione: dapsone, primaquine/tafenoquine, rasburicase, nitrofurantoin, sulfonamides, methylene blue. These generate H₂O₂/ROS that the deficient cell cannot detoxify, causing acute hemolysis.

BOARD TRAP

Rasburicase and methylene blue are absolutely contraindicated; methylene blue (the antidote for other methemoglobinemia) paradoxically worsens it in G6PD deficiency.

6. Favism — fava beans

WHAT YOU SEE

Door 2 'favism', ingestion only, broad beans

WHAT IT ENCODES

Fava (broad) beans contain vicine/convicine, potent oxidants causing acute hemolysis ('favism'), especially in Mediterranean variant children. Ingestion is the trigger; broad beans are unsafe.

BOARD TRAP

Favism is variant-dependent — classically Mediterranean, often absent in African A-. Don't assume all G6PD patients react to fava beans.

7. Infection trigger

WHAT YOU SEE

Door 3 'infection trigger'

WHAT IT ENCODES

Infection is actually the MOST common precipitant of hemolysis in G6PD deficiency — neutrophil oxidative burst generates ROS. Pneumonia, hepatitis, typhoid, and other febrile illnesses can trigger episodes.

BOARD TRAP

Students fixate on drugs/fava beans, but infection is the leading trigger overall — a frequent board point.

8. Primaquine special rule

WHAT YOU SEE

Primaquine special rule box: screen first

WHAT IT ENCODES

Always screen G6PD before primaquine/tafenoquine for malaria radical cure, since these 8-aminoquinolines cause dose-dependent hemolysis. Single low-dose primaquine (gametocytocidal) is considered safe even in deficiency per WHO.

BOARD TRAP

Screen-first for radical cure; but a single low-dose primaquine for transmission-blocking is acceptable without screening — know the distinction.

9. Self-limited episodic hemolysis

WHAT YOU SEE

Pale man 'usually self-limited, no treatment'

WHAT IT ENCODES

In the common African A- variant, hemolysis is self-limited: only older red cells (lowest enzyme) lyse, then reticulocytes with higher enzyme replace them. Supportive care and removing the trigger usually suffice.

BOARD TRAP

This self-limiting age-dependent pattern is why hemolysis abates despite continued exposure in mild variants — not seen in severe Mediterranean disease.

10. AHA: dark urine, day 2-3 peak

WHAT YOU SEE

Acute hemolytic anemia label, peak day 2-3

WHAT IT ENCODES

Acute hemolysis begins 1–3 days after trigger: dark urine (hemoglobinuria), jaundice, back/abdominal pain, with anemia peaking around day 7. Intravascular hemolysis releases free hemoglobin.

BOARD TRAP

Timing matters — a normal day-0 enzyme assay during acute hemolysis can be falsely normal; recheck after recovery.

11. Acute renal failure risk

WHAT YOU SEE

Severe transfusion, acute renal failure

WHAT IT ENCODES

Massive intravascular hemolysis can precipitate acute kidney injury from free hemoglobin and pigment nephropathy. Severe cases need transfusion and aggressive hydration.

BOARD TRAP

Free-hemoglobin AKI is a complication of severe episodes — manage with hydration; transfuse only for severe symptomatic anemia.

12. Bite cells & blister cells

WHAT YOU SEE

Bite cells / hemighosts on smear

WHAT IT ENCODES

Splenic macrophages pit denatured hemoglobin (Heinz bodies), leaving 'bite cells' and 'blister/hemighost' cells on peripheral smear — hallmark morphology of oxidative hemolysis.

BOARD TRAP

Bite cells point to oxidative injury (G6PD); spherocytes point to HS/AIHA. Don't confuse the two morphologies.

13. Heinz bodies (supravital stain)

WHAT YOU SEE

Heinz bodies label, NADPH pipeline

WHAT IT ENCODES

Heinz bodies are precipitated denatured hemoglobin seen only with supravital stains (crystal violet/new methylene blue), not on routine Wright stain. NADPH from G6PD normally keeps glutathione reduced to prevent this.

BOARD TRAP

Heinz bodies are invisible on standard smear — you must order a supravital stain to see them.

14. False-normal enzyme during crisis

WHAT YOU SEE

FALSE NORMAL box: old cells destroyed

WHAT IT ENCODES

During acute hemolysis the enzyme-deficient old cells are already lysed, and young reticulocytes have higher G6PD, so a quantitative assay can read falsely normal. Repeat testing 2–3 months after recovery.

BOARD TRAP

Classic trap: a 'normal' G6PD level during a crisis does NOT exclude deficiency — retest after the reticulocytosis resolves.

15. Quantitative assay is definitive

WHAT YOU SEE

Quantitative assay 'definitive, even during attack' caveat

WHAT IT ENCODES

Quantitative spectrophotometric G6PD assay measuring NADPH generation is the confirmatory diagnostic test. Fluorescent spot screening is a rapid alternative. Confirm timing relative to the hemolytic episode.

BOARD TRAP

Point-of-care fluorescent spot tests screen but the quantitative assay confirms — and both can mislead during active hemolysis.
